

A Practical Guide to K-nearest neighbors algorithm

TOPIC -- K-NEAREST NEIGHBORS ALGORITHM

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Properties k-NN is a special case of a variable-bandwidth, kernel density "balloon" estimator with a uniform kernel. The naive version of the algorithm is easy to implement by computing the distances from the test example to all stored examples, but it is computationally intensive for large training sets. Using an approximate nearest neighbor search algorithm makes k-NN computationally tractable even for large data sets. Many nearest neighbor search algorithms have been proposed over the years; these generally seek to reduce the number of distance evaluations actually performed. k-NN has some strong consistency results. As the amount of data approaches infinity, the two-class k-NN algorithm is guaranteed to yield an error rate no worse than twice the Bayes error rate (the minimum achievable error rate given the distribution of the data). Various improvements to the k-NN speed are possible by using proximity graphs. For multi-class k-NN classification, Cover and Hart (1967) prove an upper bound error rate of

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Error rates There are many results on the error rate of the k nearest neighbour classifiers. The k-nearest neighbour classifier is strongly (that is for any joint distribution on (X, Y)) consistent provided $k := k_n$ diverges and k_n/n converges to zero as $n \rightarrow \infty$. Let $C_{k,n}$ denote the k nearest neighbour classifier based on a training set of size n. Under certain regularity conditions, the excess risk yields the following asymptotic expansion

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$$R^* \leq R_{k,n} \leq R^* + (2 - MR^* - 1) R_{k,n} \leq R^* + \left(2 - \frac{MR^*}{M-1}\right) R_{k,n}$$

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Further reading Dasarthy, Belur V., ed. (1991). Nearest Neighbor (NN) Norms: NN Pattern Classification Techniques. IEEE Computer Society Press. ISBN 978-0818689307. Shakhnarovich, Gregory; Darrell, Trevor; Indyk, Piotr, eds. (2005). Nearest-Neighbor Methods in Learning and Vision. MIT Press. ISBN

978-0262195478.

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Feature extraction When the input data to an algorithm is too large to be processed and it is suspected to be redundant (e.g. the same measurement in both feet and meters) then the input data will be transformed into a reduced representation set of features (also named features vector). Transforming the input data into the set of features is called feature extraction. If the features extracted are carefully chosen it is expected that the features set will extract the relevant information from the input data in order to perform the desired task using this reduced representation instead of the full size input. Feature extraction is performed on raw data prior to applying k-NN algorithm on the transformed data in feature space. An example of a typical computer vision computation pipeline for face recognition using k-NN including feature extraction and dimension reduction pre-processing steps (usually implemented with OpenCV):

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Condensed Nearest Neighbor for data reduction Condensed nearest neighbor (CNN, the Hart algorithm) is an algorithm designed to reduce the data set for k-NN classification. It selects the set of prototypes U from the training data, such that 1NN with U can classify the examples almost as accurately as 1NN does with the whole data set.

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$$R(R(C_{n \times k \times n}) - R(R(C_{\text{Bayes}})) = \{B_1 \frac{1}{k} + B_2 \left(\frac{k}{n}\right)^4 / d\} \{1 + o(1)\},$$

$$\{\mathcal{R}\}_{\mathcal{R}}(C_n^{\text{knn}}) - \{\mathcal{R}\}_{\mathcal{R}}(C^{\text{Bayes}}) = \left\{B_1 \frac{1}{k} + B_2 \left(\frac{k}{n}\right)^4 / d\right\} \{1 + o(1)\},$$

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Scan all elements of X , looking for an element x whose nearest prototype from U has a different label than x . Remove x from X and add it to U . Repeat the scan until no more prototypes are added to U . Use U instead of X for classification. The examples that are not prototypes are called "absorbed" points. It is efficient to scan the training examples in order of decreasing border ratio. The border ratio of a training example x is defined as

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for some constants B_1 $\{\displaystyle B_1\}$ and B_2 $\{\displaystyle B_2\}$. The choice $k^* = \left\lfloor \frac{B_1 n^4}{d + 4 B_2} \right\rfloor$ $\{\displaystyle k^* = \left\lfloor \frac{B_1 n^4}{d + 4 B_2} \right\rfloor\}$

$\lceil \frac{4}{d+4} \rceil$ offers a trade off between the two terms in the above display, for which the k^* -nearest neighbour error converges to the Bayes error at the optimal (minimax) rate $O(n^{-\frac{4}{d+4}})$.

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random error insufficient training examples of this class (an isolated example appears instead of a cluster) missing important features (the classes are separated in other dimensions which we don't know) too many training examples of other classes (unbalanced classes) that create a "hostile" background for the given small class Class outliers with k -NN produce noise. They can be detected and separated for future analysis. Given two natural numbers, $k > r > 0$, a training example is called a (k, r) NN class-outlier if its k nearest neighbors include more than r examples of other classes.

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$w_{ni} = 0$ for $i = k^* + 1, \dots, n$. With optimal weights the dominant term in the asymptotic expansion of the excess risk is $O(n^{-\frac{4}{d+4}})$. Similar results are true when using a bagged nearest neighbour classifier.

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Data reduction Data reduction is one of the most important problems for work with huge data sets. Usually, only some of the data points are needed for accurate classification. Those data are called the prototypes and can be found as follows:

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Decision boundary Nearest neighbor rules in effect implicitly compute the decision boundary. It is also possible to compute the decision boundary explicitly, and to do so efficiently, so that the computational complexity is a function of the boundary complexity.

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$R(C_n) - R(C_{\text{Bayes}}) = (B_1 s_n^2 + B_2 t_n^2) \{1 + o(1)\}$,

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